IMRAN AHSAN

DATA SCIENCE • GRAPH MACHINE LEARNING • TRUSTWORTHY AI

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PROFESSIONAL PROFILE

AI and data-science researcher specializing in trustworthy graph machine learning, with a Ph.D. in Engineering from Chung-Ang University, six years of university teaching, and more than four years of professional software-development experience. Doctoral research focused on efficient graph unlearning, explainability-based verification, privacy diagnostics, and realistic deletion benchmarks for graph neural networks. Broader work spans reproducible evaluation, data quality and governance, information retrieval, retrieval-augmented recommender systems, and AI software engineering. Combines method development with open research artifacts, rigorous benchmarks, and implementation-oriented teaching.

RESEARCH INTERESTS

Graph Neural Networks	Graph Unlearning	Explainable AI (XAI)	Trustworthy / Responsible AI	Privacy-Preserving AI

EDUCATION

Ph.D. in Engineering | Chung-Ang University, Seoul, Republic of Korea **2023-2026**
Department of Smart Cities | Degree awarded: 21 August 2026
- Doctoral defense completed June 2026; CGPA: 4.26/4.50.
- Dissertation: "Toward Efficient and Effective Graph Unlearning: Challenges, Methods, and Evaluation."
- Supervisor: Professor Mucheol Kim.

M.S. in Software Engineering | National University of Sciences and Technology (NUST), Pakistan **2014-2017**
- Thesis: "Automated Test Case Generation from User Specifications using Natural Language Processing."
- CGPA: 3.25/4.00; Supervisor: Dr. Wasi Haider Butt.

B.S. in Software Engineering | International Islamic University Islamabad (IIUI), Pakistan **2009-2013**

RESEARCH, TEACHING & INDUSTRY EXPERIENCE

Postdoctoral Researcher | Data Intelligence Laboratory, Chung-Ang University **2026-Present**
- Leads advanced research tracks in trustworthy AI, model maintenance, algorithmic governance frameworks.
- Develop methods, benchmarks, and verification protocols for graph unlearning, explainable AI, and privacy-aware model updates.
- Build reproducible Python, PyTorch, and PyTorch Geometric pipelines spanning GCN, GAT, GraphSAGE, membership-inference diagnostics, and multi-view evaluation.
- Produce peer-reviewed publications, open-source implementations, systematic reviews, and benchmark artifacts with documented provenance.

Lecturer in Computer Science | National University of Modern Languages, Islamabad, Pakistan **2017-2023**
- Taught Object-Oriented Programming with Java, Software Engineering, Database Management Systems and SQL, Programming Fundamentals, C#/.NET, and software-development practice.
- Designed lectures, laboratories, project-based assessments, rubrics, examinations, and formative and summative feedback.
- Supervised final-year projects and supported curriculum development and student mentoring across varied programming backgrounds.

Software Developer | Pakistan **4+ years**
- Developed ASP.NET-based university portals, admissions systems, student-service modules, HR systems, and business applications.
- Worked with C#, ASP.NET, JavaScript, HTML/CSS, Microsoft SQL Server, MySQL, and Oracle in database-backed application environments.

PUBLICATIONS

Journal Articles (5)

Journal metrics use the 2025 Journal Citation Reports (released in 2026). Where a journal is assigned to multiple categories, category-specific quartiles are stated.

1. **Imran Ahsan**, Brij B. Gupta, Jinsung Kim, and Mucheol Kim. "NodeVanisher: Efficient and Scalable Data Unlearning in GNNs through Targeted Subgraph Refinement." *IEEE Transactions on Consumer Electronics*, accepted, 2026.
   Metrics: JCR 2025 Q1; Journal Impact Factor 9.9.

2. **Imran Ahsan**, Muhammad Waseem Anwar, JungYoon Kim, and Mucheol Kim. "Unlocking the Power of Graph Neural Networks-A Systematic Literature Review of Application-Oriented GNN Studies." *Computer Modeling in Engineering & Sciences*, vol. 148, no. 1, Article 3, 2026. doi:10.32604/cmes.2026.080382 | artifacts
   Metrics: JCR 2025 Q1 (Mathematics, Interdisciplinary Applications) and Q2 (Engineering, Multidisciplinary); Journal Impact Factor 2.9.

3. Minhyeok Choi, **Imran Ahsan**, Hyunwook Yu, Taeyoung Choe, and Mucheol Kim. "The Semantic Design Space of Retrieval-Augmented Recommender Systems: A Systematic Review of LLM-Based Approaches." *Computers, Materials & Continua*, vol. 88, no. 1, Article 3, 2026. doi:10.32604/cmc.2026.079504
   Metrics: JCR 2025 Q3 (Computer Science, Information Systems; Materials Science, Multidisciplinary); Journal Impact Factor 2.4.

4. Maryum Hamdani, Wasi Haider Butt, Muhammad Waseem Anwar, **Imran Ahsan**, Farooque Azam, and Mudassar Adeel Ahmed. "A Novel Framework to Automatically Generate IFML Models From Plain Text Requirements." *IEEE Access*, vol. 7, pp. 183489-183513, 2019. doi:10.1109/ACCESS.2019.2959813
   Metrics: JCR 2025 Q2 (Computer Science, Information Systems; Engineering, Electrical & Electronic; Telecommunications); Journal Impact Factor 4.2.

5. Mudassar Adeel Ahmed, **Imran Ahsan**, and Farooque Azam. "Model Driven Architecture: Business Modelling for Software Standard Models." *International Journal of Computer Science and Information Security*, vol. 14, no. 9, pp. 24-34, 2016.
   Metrics: Not listed in the 2025 JCR; JCR quartile and Journal Impact Factor are unavailable.

Conference Publications (10)

1. **Imran Ahsan**, Hyunwook Yu, Jinsung Kim, and Mucheol Kim. "Forget and Explain: Transparent Verification of GNN Unlearning." *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining (WSDM 2026)*, pp. 1063-1067, 2026. doi:10.1145/3773966.3779386 | code

2. **Imran Ahsan**, Jinsung Kim, and Mucheol Kim. "From Random Forget-Sets to Realistic Natural-Language Deletion Requests." *Proceedings of The ACM Web Conference 2026 (WWW 2026)*, pp. 8741-8744, 2026. doi:10.1145/3774904.3792954 | code

3. **Imran Ahsan**, Hyunwook Yu, and Mucheol Kim. "GNN Unlearning Reality Checklist (GURC): A Standard for Robust, Reproducible, and Privacy-Safe Evaluation." *2025 16th International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 874-876, 2025.

4. Mudassar Adeel Ahmed, **Imran Ahsan**, Usman Qamar, and Wasi Haider Butt. "A Novel Natural Language Processing Approach to Automatically Visualize Entity-Relationship Model from Initial Software Requirements." *3rd International Conference on Communication Technologies (ComTech 2021)*, pp. 39-43, 2021. doi:10.1109/ComTech52583.2021.9616949

5. Muhammad Waseem Anwar, **Imran Ahsan**, Farooque Azam, Wasi Haider Butt, and Muhammad Rashid. "A Natural Language Processing (NLP) Framework for Embedded Systems to Automatically Extract Verification Aspects from Textual Design Requirements." *12th International Conference on Computer and Automation Engineering (ICCAE 2020)*, pp. 7-12, 2020. doi:10.1145/3384613.3384619

6. **Imran Ahsan**, Mudassar Adeel Ahmed, Saad Rehman, Muhammad Abbas, and Muazzam Ali Khan. "A Novel NLP Application to Automatically Generate Text Extraction Concepts from Textual Descriptions." *Proceedings of the 2019 5th International Conference on Computing and Artificial Intelligence (ICCAI 2019)*, pp. 55-58, 2019. doi:10.1145/3330482.3330506

7. **Imran Ahsan**, Wasi Haider Butt, Mudassar Adeel Ahmed, and Muhammad Waseem Anwar. "A Comprehensive Investigation of Natural Language Processing Techniques and Tools to Generate Automated Test Cases." *Proceedings of the Second International Conference on Internet of Things, Data and Cloud Computing (ICC 2017)*, Article 132, pp. 1-10, 2017. doi:10.1145/3018896.3036375

8. Mudassar Adeel Ahmed, Wasi Haider Butt, **Imran Ahsan**, Muhammad Waseem Anwar, Muhammad Latif, and Farooque Azam. "A Novel Natural Language Processing (NLP) Approach to Automatically Generate Conceptual Class Model from Initial Software Requirements." *Information Science and Applications 2017*, Lecture Notes in Electrical Engineering, vol. 424, pp. 476-484, Springer, 2017. doi:10.1007/978-981-10-4154-9_55

9. Mudassar Adeel Ahmed, **Imran Ahsan**, and Muhammad Abbas. "Systematic Literature Review: Ingenious Software Project Management while Narrowing the Impact Aspect." *Proceedings of the 2016 Conference on Research in Adaptive and Convergent Systems (RACS 2016)*, pp. 165-168, 2016. doi:10.1145/2987386.2987422

10. Mudassar Adeel Ahmed, **Imran Ahsan**, Muazzam A. Khan Khattak, and Saad Rahman. "Management of Road Lights System Using Vehicular Ad-Hoc Networks." *2016 2nd IEEE International Conference on Computer and Communications (ICCC 2016)*, pp. 2292-2296, 2016. doi:10.1109/CompComm.2016.7925108

Under Review/Submitted/In Preparation

1. **Imran Ahsan** et al. "GDPRBench: A Controller-Side Benchmark for Accuracy-Triggered Graph Unlearning." *In preparation*, 2026.

2. **Imran Ahsan** et al. "Retrain-Aligned Explanation Auditing for GNN Unlearning." *In preparation, 2026*.

RESEARCH CONTRIBUTIONS & OPEN SCIENCE

- Efficient graph unlearning: request-centered and subgraph-local update strategies that reduce retraining scope while preserving model utility and privacy-oriented evaluation.
- Transparent verification: explanation-shift and structural diagnostics that provide human-understandable evidence of how designated graph entities influence a model before and after unlearning.
- Realistic and auditable evaluation: natural-language erasure requests, data-quality-triggered forget sets, reproducible provenance, and multi-view assessment protocols for graph learning systems.
- Evidence synthesis and research software: systematic reviews of application-oriented GNNs and retrieval-augmented recommendation, supported by public implementations and reusable benchmark artifacts.

Open resources: F-E | Text2Forget | GNN-SLR

TEACHING & SUPERVISION

- Courses: Object-Oriented Programming with Java; Software Engineering; Database Management Systems and SQL; Programming Fundamentals; C#/.NET; and software-development practice.
- Instructional design: lecture-laboratory integration, project-based assessment, rubric development, examination design, marking, and formative and summative feedback.
- Student development: final-year project supervision and mentoring in software design, database-backed applications, testing, implementation, and technical documentation.

TECHNICAL EXPERTISE

Graph AI & Trustworthy ML	Graph neural networks; graph and machine unlearning; explainable AI; privacy diagnostics; responsible model updating; data-quality-aware evaluation.
Data Science & Research	Experimental design; benchmark construction; reproducible evaluation; systematic literature reviews; natural language processing; information retrieval; retrieval-augmented systems.
Programming & ML	Python; PyTorch; PyTorch Geometric; scikit-learn; Java; C#; ASP.NET; JavaScript.
Databases & Web	SQL; Microsoft SQL Server; MySQL; Oracle; database-backed enterprise applications; HTML/CSS.

HONORS & SCHOLARSHIPS

- Chung-Ang University Young Scientist Scholarship (CAYSS), 2023.
- Brain Korea 21 (BK21) Scholarship, 2023.
- Recommended by Chung-Ang University for the F-2-7S Global Talent Visa, 2026.

REFERENCES

1. **Professor Mucheol Kim** | Data Intelligence Laboratory, Chung-Ang University, Seoul, Republic of Korea
2. **Professor Wasi Haider Butt |** National University of Science and Technology, Islamabad, Pakistan

Additional references are available on request.